Chorn Chhery

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SUMMARY

IT Programmer at MinebeaMitsumi. I build the internal programs a factory uses (a leave system, a factory website, monitoring and a production counter) and keep them running. Earlier experience as an AI Engineer (RAG chatbots) and Full Stack Developer.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, C#, Java, C, C++, Golang

Web & Frameworks: React, NestJS, Node.js, .NET Core, Blazor WebAssembly, SignalR, Flutter (Dart), Tailwind CSS, HTMX, Bootstrap, HTML, CSS

Databases: PostgreSQL, Prisma, Redis, MySQL, Microsoft SQL Server, MongoDB, Qdrant, SQLite

DevOps & Systems: Docker, Linux (Ubuntu), nginx, Grafana, Prometheus, GitHub Actions, Git, GitHub, Azure DevOps, Raspberry Pi

AI & Other: RAG, NLP, Ollama, Telegram Bot API, Scikit-learn, Spark, Streamlit, Postman, Figma, xUnit

Languages: Khmer (Native), English (Intermediate), Thai (Advanced)

EDUCATION

Prince of Songkla University, Thailand

Bachelor of Mathematics and Computer Science | Major: Data Science | Cumulative GPA: 3.68/4.00 (First Class Honors)

PROFESSIONAL EXPERIENCE

IT Programmer – MinebeaMitsumi Inc. (Phnom Penh, Cambodia) May 2026 - Present

Build and run the internal programs our factory teams use, from leave requests to production lines.

- Built a leave system on Telegram in English and Khmer. Staff write their leave like a normal message; the program checks it, blocks duplicates and asks an admin to approve. If no admin answers in 5 hours, it approves the request by itself.
- Built a factory website that shows each production line, who is at work and who is on leave. Supervisors move workers between lines and the workers return automatically. Approved leave comes from the Telegram system every minute, so nobody types it twice.
- Added a retry queue, health checks, backups and updates that roll back, after leave requests were lost once when the database stopped.
- Set up Grafana dashboards for server health and speed, and built a Raspberry Pi counter that compares production output with the plan for each shift.

AI Engineer (Co-op Intern, promoted to Staff) – Max Savings (Bangkok, TH) August 2025 - April 2026

Built a RAG chatbot that answers employees' questions from company documents.

- Built the document pipeline: PDF upload, 3 chunking methods, and embeddings stored in Qdrant or SQLite.
- Built hybrid search (vector + BM25 with Thai tokenization, merged with RRF) across all active documents, with a 70% confidence threshold and cited sources.
- Added Redis caching and question detection, cutting response time from about 850 ms to under 100 ms.
- Built an admin console that tests chunking and embedding setups and scores answers with BLEU, GLEU, F1 and star ratings; added RAG version control (pause, roll back).
- Built the chat app in Blazor WebAssembly and SignalR: real-time chat, login with roles, chat history, per-user settings, dark mode.
- Wrote 30+ REST endpoints, xUnit tests, and user and API manuals. As co-op intern: upgraded the project from .NET 9 to .NET 10, designed the UI in Figma, and built an email screen and QR-code check-in.

Full Stack Developer Intern – Biomass Exchange (Bangkok, TH) February 2025 - May 2025

Built the email notification system in Go for BiomassX, a bioenergy exchange.

- Built 14 automated emails in Thai and English (28 templates) for registration, orders, matches, contracts, rejection, cancellation and reminders.
- Added a contract flow and scheduled jobs: hourly retry of pending emails, a day-3 reminder, and automatic cancellation after 3 days.
- Sent emails in the background (goroutines, 5 workers) so pages respond instantly; each email is saved as pending, sent or failed.
- Designed the PostgreSQL notifications table, and built a booking history feature with full CRUD (HTMX, JavaScript, CSS).

PROJECTS

Portfolio website: chornchhery.github.io (more projects, with screenshots and details)

ChatBotRAG – Document Question Answering Chatbot: github.com/ChornChhery/RAG-

- Full-stack RAG chatbot using Blazor WebAssembly, .NET Core and SignalR, with token-by-token streaming and local Ollama models for full privacy.
- Hybrid search (Vector + BM25) with tokenization for English, Thai and Khmer documents.
- Automated answer scoring with BLEU, GLEU, F1 and an LLM judge, so no manual reference answers are needed.

Real-Time IoT Poultry Monitoring System: github.com/ChornChhery/Chicken-Farm-IOT

- Real-time monitoring with automatic data ingestion into MongoDB to track poultry growth.
- Data validation pipeline that filters sensor readings, reaching 95% data accuracy.
- Live dashboard and automatic LINE notifications for growth problems.

LEADERSHIP EXPERIENCE & ACTIVITIES

International English Exchange Program (Universiti Malaysia Kelantan), Malaysia 8 - 14 September 2025

- Improved English communication through international workplace training.
- Led group discussions and team projects to build understanding and teamwork.
- Built international relationships while representing Cambodian culture.

Technical Assistant – Prince of Songkla University, Hat Yai Campus 13 - 15 December 2024

- Supported Python and robotics workshops for 50+ students at a Programming Camp.
- Helped participants with debugging, problem-solving and logic.